9 September 1976
SUPERSEDING
(See 6.4)

MILITARY SPECIFICATION

# DISPENSER, BUTTER, SELF CONTAINED, EUTECTIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a eutectic type butter dispenser furnished with two eutectic cooling cartridges.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard

MILITARY

MIL-P-116 - Preservation-Packaging, Methods Of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Natick Research and Development Command, ATTN: DRXNM-TDG, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7320

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MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U.S. Military Property

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL SANITATION FOUNDATION (NSF)

Criteria C-2 - Special Equipment and/or Devices

Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, P.O. Box 1468, Ann Arbor, MI 48106.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)



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## 3. REQUIREMENTS

3.1 Standard product. The butter dispensers shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements of this specification. Modifications to add features shall not incorporate different parts unless such parts are used on other current commercial models. Standard or modified commercial products furnished in accordance with this specification shall be identifiable by all regular manufacturer's or commercial service organizations servicing the brand involved. Service organizations shall be capable of providing complete parts and repair services on models furnished to the Government consistent with their normal commercial practices.

3.2 Codes and standards. The dispenser shall comply with the applicable requirements of NSF Criteria C-2.

3.2.1 Compliance. Prior to approval of the first shipment, the contractor shall submit to the contracting officer or his authorized representative satisfactory evidence that the dispenser he proposes to furnish under this specification meet the applicable requirements of NSF Criteria C-2 as follows.

3.2.1.1 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF seal on the finished dispenser, or

(2) A certification for the dispenser issued by NSF under their special one-time contract evaluation/certification service, or

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the dispensers are being procured, indicating that the dispensers have been tested and conform to NSF.

3.3 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial dispensers provided the completed item complies with all provisions of this specification.

3.3.1 Stainless steel. Stainless steel sheet and strip shall conform to type 302 or 304 of ASTM A 167.



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3.4 Design. The dispenser shall be an eutectic cooled cabinet type for countertop use that dispenses individual butter pats by manual extraction by hand. The dispenser shall be sized to accommodate a minimum of 4 prepacked trays of butter pats 1-1/4 inch (131.8 mm) square placed on plastic or bleached paper board chips covered with waxed or parchment paper. Butter tray loading shall be accomplished without the use of tools. Butter pats shall be visible through a tempered glass or acrylic front panel.

3.4.1 Cabinet. The dispenser cabinet shall be fabricated of stainless steel specified in 3.3.1. The cabinet shall be provided with an easily cleanable solid bottom or tray beneath the butter dispensing area. The maximum outside dimensions of the dispenser shall be 14-1/2 inches (368 mm) high by 13 inches (330 mm) wide by 11-1/2 inches (292 mm) deep.

3.4.2 Eutectic cooling units. Each eutectic cooling unit shall be capable of maintaining butter temperature between 38F (3C) to 45F (7C) in a minimum 90F (32C) ambient and 85 ± 5 percent relative humidity for 4 hours when tested as specified in 4.4.1. Each butter dispenser shall be furnished with two eutectic cooling units.

3.5 Finish. The finish shall conform to NSF Criteria C-2.

3.6 Marking. Each dispenser shall be permanently and legibly marked with the manufacturer's name, model and serial number, trade name, or trademark of such known character as to be readily identifiable to the manufacturer. Marking shall be applied in accordance with MIL-STD-130.

3.7 Workmanship. All components and assemblies of the dispenser shall be free from dirt and other harmful extraneous materials, burrs, slivers, rough die, tool and grind marks, dents and cracks. External surfaces shall be free from sharp edges and corners, except when sharp corners are required.

3.7.1 Metal fabrication. Metal used in the fabrication of the dispenser shall be free from kinks and unspecified bends. Forming and shearing shall not cause damage to the metal and shall be done neatly and accurately. Corners shall be square and true and of uniform size and shape.

3.7.2 Threaded fasteners. Threaded fasteners and rivet holes shall be accurately punched or drilled and shall have burrs removed. Threaded fasteners shall not be broken, cracked or stripped and shall be drawn tight. Rivets shall fill the holes completely and the heads shall be in full contact with the surface of the member and concentric with the hole.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract,



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the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 End product inspection. The lot shall be all dispensers offered for inspection at one time. The sample unit shall be one completely fabricated dispenser.

4.2.2.1 Visual examination. The dispensers shall be examined for the defects in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of visual defects

| Examine | Defect | Classification Major | Minor |
|---|---|---|---|
| Finish | Not as specified | X | |
| | Rough or sharp edges, burr(s) slivers, or scratches | X | |
| Design | Not adequate for countertop use | X | |
| | Butter pats cannot be extracted by hand | X | |
| | Tools required for loading dispenser | X | |
| | Butter pats not visible | X | |
| | Easily cleanable solid bottom or tray under butter dispensing area not provided | X | |
| | Two eutectic units not furnished | X | |



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TABLE I. Classification of visual defects (cont'd)

| Examine | Defect | Classification Major | Minor |
|---|---|---|---|
| Construction and workmanship | Any burr, sliver, sharp edge, rough die, tool gauge, trim or grind marks | X | |
| | Kinks or unspecified bends | X | |
| | Fractured, buckled, punctured, dented, malformed, or damaged | X | |
| | Thread fasteners broken, cracked, stripped thread, crossed thread | X | |
| | Thread fasteners loose | X | |
| | Rivet heads and peened ends not in full contact with joined metal surfaces | X | |
| Marking identification | Missing, incomplete, or illegible | | X |

4.2.2.2 Dimensional examination. The dispenser shall be examined for compliance with dimensions specified. Any dimension not within the specification tolerance shall be a defect. The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

4.2.2.3 End item testing. The test specified in 4.4.1 shall be performed on one production unit (see 6.3). Failure of the test shall be cause for rejection of the inspection lot.

4.2.3 Examination of packaging. An examination shall be made to determine that preservation-packaging, packing and marking comply with the requirements of section 5. Defects shall be as specified in table II. The sample unit shall be one shipping container fully packaged except that it need not be closed. The lot shall be the number of shipping containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 2.5 defects, expressed in terms of defects per hundred units.

TABLE II. Examination of packaging

| Examine | Defect |
|---|---|
| Marking (exterior) | Omitted; incorrect; illegible; of improper size, location, sequence, or method of application. |
| Materials | Component missing, damaged, or otherwise defective. |



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TABLE II. Examination of packaging (cont'd)

| Examine | Defect |
|---|---|
| Workmanship | Buldged or distorted containers. |
| Weight | Weight of contents per shipping container is more than required. |

4.2.3.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing and reinforcing shall be in accordance with the appendix of PPP-B-636.

4.3 Certificate examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the requirements shall be examined and validated.

4.4 Test methods.

4.4.1 Performance. The test shall be performed in a minimum ambient temperature of 90F (32C) and a relative humidity of 85 ± 5 percent. The eutectic cooling unit shall be refrigerated at 0F (-18C) ± 2F (1C) for 8 hours. The prepackaged butter trays shall be refrigerated at 38F (3C) ± 2F (1C). Load the dispenser with refrigerated butter trays and eutectic cooling unit. At the end of 4 hours the butter temperature in the middle and 1 inch (25 mm) from the top of each tray shall be recorded to determine compliance with 3.4.2. Any nonconformance with specified requirements shall constitute failure of this test.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each dispenser with cartridges shall be preserved-packaged in accordance with method III of MIL-P-116.

5.1.2 Level C. Each dispenser with cartridges shall be preserved-packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The package shall be the same as that normally used by the contractor for retail distribution.



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5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Dispensers with cartridges, preserved-packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each fiberboard shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636. The weight of contents of each fiberboard container shall not exceed 65 pounds (29.5 kg).

5.2.2 Level B. Dispensers with cartridges, preserved-packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each fiberboard container shall be closed in accordance with method II as specified in the appendix of PPP-B-636. The weight of contents of each fiberboard container shall not exceed 65 pounds (29.5 kg).

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636, and closed in accordance with method III as specified in the appendix thereto.

5.2.3 Level C. Dispensers with cartridges, preserved-packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The dispenser is intended for use in feeding lines to maintain butter pats at chilled temperatures.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation-packaging, and packing (see 5.1 and 5.2).
(c) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).



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6.2.1 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

6.3 End item test. Performance of the test specified in 4.4.1 shall be limited to one successful demonstration of compliance to the requirements of this specification for each contract.

6.4 Supersession data. This specification includes the requirements of US Army Natick Research and Development Command interim purchase description IP/DES S-1-6, dated 9 January 1976.

Custodians:

Army - GL
Navy - YD
Air Force - 99

Preparing activity:

Army - GL

Project No. 7320-0632

Review activity:

Army - MD

User activities:

Navy - MC, SA



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DOCUMENT IDENTIFIER AND TITLE
DISPENSER, BUTTER, SELF CONTAINED, EUTECTIC MIL-D-43953

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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TELEPHONE NO.

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